Certifying Functional Dependencies over Latent Entity Identity: Proofs

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Technical report accompanying the ICDE 2027 submission. Code: `https://github.com/suanlab/SemFD`

Abstract

We give full proofs for the results of the paper *Certifying Functional Dependencies over Latent Entity Identity*. Value equality is defined by an oracle that maps each value to its entity, and a functional dependency is measured by its g_3 error on the instance that the oracle induces. We prove the soundness of the repair-witness certificate (with an explicit miss-mass term, which vanishes under inverse oracle access), its validity under data-dependent stopping and its per-sample oracle cost, a lower bound for certifiers without inverse access, certified closure under Armstrong's rules, and the failure of the pair-conditional measure g_1^c under inference. We also record the protocol of the LLM-oracle experiment.

1 Setting

A relation r is a finite multiset of tuples over attributes $\mathcal{A}$. For each $A \in \mathcal{A}$ an *oracle* is a function $f_A : \mathrm{dom}(A) \to \mathcal{E}_A$; $a \equiv_A b$ iff $f_A(a) = f_A(b)$. The canonical instance $\hat{r}$ replaces each cell $t[A]$ by $f_A(t[A])$. For single attributes X, Y, $g_3(X \to Y) = \min\{|S|/|r| : r \setminus S \models X \to Y\}$, evaluated on $\hat{r}$ unless stated otherwise. Tuples are drawn uniformly at random (with replacement) from r.

Proposition 1 (Transitivity by construction)**.** *$\equiv_A$ is an equivalence relation. For a symmetric reflexive relation J on $\mathrm{dom}(A)$, the statements below hold with $\equiv_A$ replaced by the transitive closure J^*, and in general not for J itself.*

Proof. $\equiv_A$ is the kernel of a function, hence reflexive, symmetric and transitive. The proofs below use only these three properties of $\equiv$, which J^* has. Lemma 1 uses transitivity essentially, and the counterexample $a\,J\,b$, $b\,J\,c$, $\neg(a\,J\,c)$ breaks the chain argument. □

Definition 1 (Repair witness)**.** *$W = (\pi, rep, N, \prec)$ consists of*

- *a partition π of $\mathrm{dom}(X)$ into clusters;*
- *a map rep from clusters to $\mathrm{dom}(Y)$;*
- *for each $a \in \mathrm{dom}(X)$ a set $N(a) \subseteq \mathrm{dom}(X)$;*
- *a strict total order $\prec$ on clusters.*

Clusters c, c' conflict *if $rep(c) \not\equiv_Y rep(c')$. A tuple t with $a = t[X]$, $c = \pi(a)$ is* kept*, written $t \in \mathcal{K}(W)$, iff* (i) *$t[Y] \equiv_Y rep(c)$ and* (ii) *there is no $b \in N(a)$ with $\pi(b) \succ c$, $b \equiv_X a$ and $\pi(b)$ conflicting with c. Let $Z(t)$ hold iff some $b \in \mathrm{dom}(X) \setminus N(a)$ satisfies $\pi(b) \succ c$, $b \equiv_X a$ and $\pi(b)$ conflicts with c. Define $\nu(N) = \Pr_t[Z(t)]$.*

2 Soundness

Lemma 1. *$K' = \{t \in \mathcal{K}(W) : \neg Z(t)\}$ satisfies $X \to Y$ on $\hat{r}$.*

Proof. Let $t_1, t_2 \in \mathcal{K}'$ with $t_1[X] \equiv_X t_2[X]$, and let $c_i = \pi(t_i[X])$. We show $t_1[Y] \equiv_Y t_2[Y]$. *Case $c_1 = c_2$.* By (i), $t_1[Y] \equiv_Y rep(c_1) \equiv_Y t_2[Y]$, and transitivity gives the claim. *Case $c_1 \neq c_2$ and the clusters do not conflict.* Then $t_1[Y] \equiv_Y rep(c_1) \equiv_Y rep(c_2) \equiv_Y t_2[Y]$. *Case $c_1 \neq c_2$ and the clusters conflict.* W.l.o.g. $c_2 \prec c_1$. Put $b = t_1[X]$: then $\pi(b) = c_1 \succ c_2$, $b \equiv_X t_2[X]$, and c_1 conflicts with c_2. If $b \notin N(t_2[X])$ then $Z(t_2)$ holds, contradicting $t_2 \in \mathcal{K}'$. Otherwise (ii) fails for t_2, contradicting $t_2 \in \mathcal{K}(W)$. So this case cannot occur. □

Theorem 1 (Soundness)**.** *For every witness W: $g_3(X \to Y) \leq \Pr_t[t \notin \mathcal{K}(W)] + \nu(N)$.*

Proof. By Lemma 1, removing $S = r \setminus \mathcal{K}'$ yields an instance satisfying $X \to Y$. Hence $g_3 \leq |S|/|r| \leq \Pr[t \notin \mathcal{K}(W)] + \Pr[Z(t)]$. □

Corollary 1 (Inverse access)**.** *If $N(a) \supseteq f_X^{-1}(f_X(a)) \cap \mathrm{dom}(X)$ for every a, then $\nu(N) = 0$ and $g_3 \leq \Pr[t \notin \mathcal{K}(W)]$.*

Proof. Every $b \equiv_X a$ lies in $N(a)$, so $Z(t)$ never holds. □

Remark 1 (Composite left-hand sides)**.** *For $X = \{A, B\}$, let $f_X(a, b) = (f_A(a), f_B(b))$. This is a function, so all statements above apply verbatim with $\mathrm{dom}(X)$ the set of value pairs occurring in r. The inverse is $f_X^{-1}(e_A, e_B) \cap \mathrm{dom}(X) = \{(a, b) \in \mathrm{dom}(X) : a \in f_A^{-1}(e_A),\, b \in f_B^{-1}(e_B)\}$, so complete component inverses give a complete composite inverse and Corollary 1 applies. Calls are charged through the component caches.*

3 Estimation and cost

Let $B(t) = \mathbb{1}[t \notin \mathcal{K}(W)] \in \{0, 1\}$ and $\mu = \mathbb{E}[B]$. Draw $t_1, t_2, \ldots$ i.i.d. uniform. Let (L_n, U_n) be the predictable-plug-in empirical-Bernstein confidence sequence of Waudby-Smith and Ramdas [6] (Thm. 2 of arXiv:2010.09686v7), with running intersection, where each side is a one-sided sequence at level α; in particular $\Pr[\exists n : \mu \geq U_n] \leq \alpha$.

Theorem 2 (Certification)**.** *Fix δ, a declared $\bar{\nu} \geq 0$ and a cap $n_{\max}$. Stop at the first $n \leq n_{\max}$ with $U_n < \delta - \bar{\nu}$ (output* CERTIFIED*) or with $L_n \geq \delta - \bar{\nu}$, or at $n_{\max}$ (output* NOT CERTIFIED*). Then:*

1. *(Validity) If $\nu(N) \leq \bar{\nu}$, then $\Pr[\text{CERTIFIED} \wedge g_3 > \delta] \leq \alpha$. Only the upper sequence carries the guarantee, so it is used one-sided at level α (with $\log(1/\alpha)$ in place of $\log(2/\alpha)$). The lower sequence serves only to stop early for futility, which cannot affect validity. A* NOT CERTIFIED *output carries no guarantee. Under inverse access the condition holds with $\bar{\nu} = 0$.*

2. *(Per-sample cost) Evaluating $B(t)$ costs at most $3 + 3\kappa$ oracle calls (fewer with caching), where $\kappa = \max_a |N'(a)|$ and $N'(a) = N(a)$, or $N(a) \cup (f_X^{-1}(f_X(a)) \cap \mathrm{dom}(X))$ under inverse access: two Y-calls, at most one inverse call, and at most three calls per candidate.*

3. *(Total cost) The total is at most $n_{\max}(3+3\kappa)$ calls. This is independent of $|r|$; it is independent of $|\mathrm{dom}(X)|, |\mathrm{dom}(Y)|$ whenever κ is, i.e. for a fixed number of proxy neighbours and alias classes of bounded size.*

Proof. (1) On the event $\mu < U_n$ for all n, which has probability $\geq 1-\alpha$ by the one-sided guarantee, certification at n gives $g_3 \leq \mu + \nu < U_n + \bar{\nu} < \delta$ by Theorem 1. Since the confidence sequence is time-uniform, the data-dependent stopping time does not affect this. (2) Condition (i) needs $f_Y(t[Y])$ and $f_Y(rep(c))$: 2 calls. Under inverse access, one call returns the alias class of $f_X(a)$. Condition (ii) examines each $b \in N'(a)$. Clusters, priorities and representatives are known without the oracle, so only candidates with $\pi(b) \succ c$ and $rep(\pi(b)) \neq rep(c)$ as strings need calls: $f_X(a)$, $f_X(b)$ and $f_Y(rep(\pi(b)))$, at most 3 per neighbour. Representatives that are equal as strings are oracle-equal. (3) Follows from (2). □

Remark 2 (Dependence on the gap)**.** *If $\mu = \delta - \bar{\nu} - \Delta$, certification requires $U_n < \delta - \bar{\nu}$. The width of the confidence sequence is $\sigma\sqrt{2\log(1/\alpha)\log n/n}\,(1+o(1))$ (one-sided) with $\sigma^2 \leq \mu(1-\mu)$ under the default λ schedule, so $n = \tilde{O}\big(\mu\log(1/\alpha)/\Delta^2\big)$ samples suffice, where $\mu = \mathbb{E}[B]$. This explains the empirical power curve, where power collapses as $\Delta \to 0$. We use it only as a heuristic, because a rigorous expected-stopping-time bound requires a stitched or LIL-type sequence.*

Remark 3 (Relation to Cormode et al. [1])**.** *With $\equiv$ observed, the audit (ii) is unnecessary, and $\Pr[t \notin \mathcal{K}(W)]$ is the standard modal-representative estimator of g_3 for a fixed partition. Our contribution is that the partition comes from an unreliable proxy. Without (ii) the estimator is unsound: two oracle-equal values in different clusters with conflicting representatives hide a violation from every per-cluster check.*

4 Limits without inverse access

A *forward-only* certifier may read r and any proxy information, and may call f_X and f_Y on values of its choice (adaptively, possibly randomized), but has no inverse access. It is *α-valid* if on every instance with $g_3(X \to Y) > \delta$ it outputs CERTIFIED with probability at most α.

Theorem 3 (Lower bound)**.** *Let $D \geq 4$, $1/D \leq \delta < 1/4$ and $m = \lfloor \delta D \rfloor + 1$ (so $m \leq 2\delta D$). Assume the oracle answers with* opaque labels*: fresh labels drawn uniformly at random without replacement from a large label space (so pairwise distinct), unknown to the certifier, i.e. identifiers that carry no information beyond equality; probabilities are over the certifier's randomness and the labels. There is an instance I_0 with $|r| = |\mathrm{dom}(X)| = D$ and $g_3(X \to Y) = 0$ such that every forward-only α-valid certifier making at most q oracle calls outputs* CERTIFIED *on I_0 with probability at most $\alpha + q^2 m/(D(D-1))$, where this and every probability below is over the certifier's internal randomness, the random labels and (in the proof) the random pair set M. Hence certifying I_0 with probability $\beta > \alpha$ requires*

$$q \;\geq\; \sqrt{(\beta-\alpha)\,D(D-1)/m} \;\geq\; \sqrt{(\beta-\alpha)(D-1)/(2\delta)}.$$

This holds for any proxy information that is independent of which values are oracle-equal.

Proof. Let $r = \{(v_i, w_i)\}_{i \leq D}$ with distinct strings v_i and distinct Y-values w_i, and let f_Y be the identity. Oracle answers are generated lazily: the oracle keeps an infinite stream of fresh, pairwise distinct random labels.

- In I_0 every newly queried value receives the next fresh label. All values are distinct entities, so $g_3 = 0$.
- In I_M, for a set M of m disjoint pairs, a newly queried value receives its partner's label if its partner was queried before, and the next fresh label otherwise. Each pair is one entity with two distinct Y-values, so $g_3(I_M) = m/D > \delta$.

Draw M uniformly at random among all sets of m disjoint pairs, independently of everything else. Both instances get the same table and the same proxy information.

Run the certifier on both instances with the same internal randomness and the same label stream. Let E be the event that the run on I_M queries a value whose partner it queried earlier. Until E occurs, both runs receive the same fresh labels in the same order, so they ask the same queries and agree in every step. In particular, if E occurs, the query set Q_0 of the run on I_0 contains both endpoints of some pair of M.

The answers on I_0 do not depend on M, so Q_0 is independent of M. Each of the at most $\binom{q}{2}$ pairs inside Q_0 belongs to M with probability $m/\binom{D}{2}$, by symmetry, so $\Pr[E] \le \binom{q}{2}\frac{2m}{D(D-1)} \le \frac{q^2 m}{D(D-1)}$. Off E the two outputs coincide. Hence $\Pr_{I_0}[\textsc{cert}] \le \mathbb{E}_M \Pr_{I_M}[\textsc{cert}] + \Pr[E] \le \alpha + \Pr[E]$, where both sides average over the certifier's randomness and the label stream (and the left side does not depend on M), using α-validity on every I_M. □

Remark 4 (Scope)**.** *The opaque-label assumption matters. An oracle that answers with a canonical name that itself occurs in the data, as an LLM answering "Walmart" may, partially reveals the inverse, and the bound need not apply.*

Remark 5. *With inverse access the same instances are easy. $f_X^{-1}(f_X(v_i))$ reveals the partner of v_i, so for any partition that does not merge distinct entities (e.g. singletons) we have $B \equiv 0$ on I_0, and Theorem 2 certifies with $O(\log(1/\alpha)/\delta)$ samples, independently of D. Inverse access, or an assumption about the proxy's recall that cannot be checked cheaply, is therefore what makes the cost independent of the domain size. Without either, the cost must grow at least like $\sqrt{D}$. The bound is modest at moderate scale: about* 550 *calls for $D =$* 29,894 *(GeoNames-16k city names), $\delta = 0.05$, $\beta - \alpha = 1/2$.*

In the hard instances the proxy cannot see the matching. This is the situation of world-knowledge aliases with no string similarity, such as Holland/Netherlands *or* Equate/Walmart*.*

The bound concerns oracle calls, not row samples. It complements the $\Omega(\sqrt{N})$ row-sampling bound of Kivinen and Mannila [3] and Cormode et al. [1], which assumes that equality is observed.

5 Certified closure

Lemma 2 (Subadditivity; Väänänen [5], rules A4/A6; see also [2])**.** *If φ is derivable from Σ' by Armstrong's rules, then on any instance $g_3(\varphi) \le \sum_{\sigma\in\Sigma'} g_3(\sigma)$.*

Proof. (Equivalently: monotonicity of the minimum-repair measure [4] plus a union of repairs.) Let S_σ be an optimal removal set for σ. Exact FDs are preserved under taking subinstances, so $r \setminus \bigcup S_\sigma$ satisfies every $\sigma \in \Sigma'$ and hence φ, by soundness of Armstrong's rules. □

Theorem 4 (Certified closure)**.** *Fix, before any sampling, a candidate set $\mathcal{C}$ and levels α_σ for $\sigma \in \mathcal{C}$, and assume $\nu_\sigma(N) \le \bar{\nu}_\sigma$ for every $\sigma \in \mathcal{C}$ (automatic under inverse access, with $\bar{\nu}_\sigma = 0$). Run the certificate on every $\sigma \in \mathcal{C}$ and let $\Sigma \subseteq \mathcal{C}$ be the certified ones, with final bounds U_σ. Then with probability $\ge 1 - \sum_{\sigma\in\mathcal{C}} \alpha_\sigma$, simultaneously for all φ and $\Sigma' \subseteq \Sigma$ with $\Sigma' \vdash \varphi$, we have $g_3(\varphi) \le \sum_{\sigma\in\Sigma'}(U_\sigma + \bar{\nu}_\sigma)$.*

Proof. By the union bound over $\mathcal{C}$, which is fixed in advance, all confidence sequences cover their means at all times with probability $\ge 1 - \sum_{\mathcal{C}} \alpha_\sigma$. On this event each $\sigma \in \Sigma$ has $g_3(\sigma) \le U_\sigma + \bar{\nu}_\sigma$ by Theorem 1, whatever data-dependent subset Σ turns out to be. Lemma 2 then bounds every derivation. Summing the α_σ only over the certified Σ would be invalid, because Σ is selected by the data. No independence between certificates is needed: they may share data and oracle caches. □

Proposition 2 (Unary supports)**.** *For unary Σ and unary $\varphi = X \to Y$ with $X \neq Y$, every support of φ contains a directed path from X to Y in the graph with edges Σ. Hence the minimum of $\sum_{\sigma \in P} U_\sigma$ over such paths, which a shortest-path algorithm computes, equals the minimum over all supports and is a valid bound. It need not be tight to $g_3(\varphi)$.*

Proof. A path is a transitivity derivation whose support is its edge set. Conversely, the attribute closure of $\{X\}$ under a set of unary FDs is the set of vertices reachable from X, so any support deriving Y contains such a path; since weights are nonnegative, dropping the other premises cannot increase the sum. Additive bounds are not tight [2], so the bound need not equal $g_3(\varphi)$. □

6 Predictable witnesses, updates and oracle changes

Proposition 3 (Predictable witnesses)**.** *Let the witness W_n used for the n-th sample be a function of the earlier samples and of the oracle answers obtained before it, and let $\mu_n = \mathbb{E}[B_n \mid \mathcal{F}_{n-1}]$ be the conditional mean of $B_n = \mathbb{1}[t_n \notin \mathcal{K}(W_n)]$. Under inverse access, $\mu_n \geq g_3$ for every n, and with probability at least $1 - \alpha$, simultaneously for all n, $U_n > \sum_{i \leq n} \lambda_i \mu_i / \sum_{i \leq n} \lambda_i \geq g_3$. Hence Theorem 2(1) holds for the self-refining witness of the system (every entity whose alias class an earlier batch fetched becomes its own cluster).*

Proof. Given $\mathcal{F}_{n-1}$, W_n is fixed and t_n is uniform, so μ_n is the $\mathbb{E}[B]$ of a fixed witness and $\mu_n \geq g_3$ by Theorem 1 with $\nu = 0$. The predictable-plug-in empirical-Bernstein process of Waudby-Smith and Ramdas [6], $M_n = \prod_{i \leq n} \exp\big(\lambda_i(\mu_i - B_i) - 4(B_i - \hat{\mu}_{i-1})^2 \psi_E(\lambda_i)\big)$, with predictable $\lambda_i \in [0, 1)$ and $\hat{\mu}_{i-1}$, satisfies $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] \leq M_{n-1}$: the one-step inequality in their proof of Theorem 2 uses only that $B_i \in [0, 1]$ has conditional mean μ_i given $\mathcal{F}_{i-1}$, not that the means are equal. Ville's inequality gives $\Pr[\exists n : M_n \geq 1/\alpha] \leq \alpha$, and on the complement, rearranging $\log M_n < \log(1/\alpha)$ gives $\sum_i \lambda_i \mu_i / \sum_i \lambda_i < U_n$ for all n. A weighted average of values $\geq g_3$ is $\geq g_3$, and the running minimum of U_n inherits the bound. □

Proposition 4 (Updates)**.** *Let r' arise from r ($|r| = N$) by deleting D and inserting I tuples, with the oracle unchanged. Then $g_3^{\hat{r}'} \leq 1 - ((1 - g_3^{\hat{r}})N - D)/(N - D + I)$, and the right-hand side is nondecreasing in $g_3^{\hat{r}}$.*

Proof. An optimal repair of r keeps $(1 - g_3^{\hat{r}})N$ tuples satisfying $X \to Y$ on $\hat{r}$. At least $(1 - g_3^{\hat{r}})N - D$ of them are in r'; their canonical values are unchanged and FDs are closed under subsets, so they form a consistent subset of r' of that size. □

Substituting $U + \bar{\nu}$ for $g_3^{\hat{r}}$ yields a bound that holds on the event of the original certificate for every later version. The bound stays below δ while $(\delta - U - \bar{\nu})N \geq \delta D + (1 - \delta)I$. Re-certifying at version j uses a level α_j fixed in advance (we use $\alpha/2^{j+1}$ divided by the number of candidates), so all versions are covered with probability at least $1 - \sum_j \alpha_j$. Forward oracle answers stay valid across versions, but alias lists do not, because insertions add spellings: a re-run must fetch the alias class of every entity it touches again.

Proposition 5 (Oracle transfer)**.** *For oracles f, f', let m be the smallest fraction of tuples whose removal leaves, for each attribute of $X \cup Y$, a value set that f and f' partition identically. Then $|g_3^{\hat{r}_f} - g_3^{\hat{r}_{f'}}| \leq m$. For one attribute, $m = 1 - \max_M \sum_{(e,e') \in M} w(e, e')$ over matchings M between f-classes and f'-classes, where $w(e, e')$ is the fraction of tuples whose value lies in $e \cap e'$.*

Proof. The first claim: remove an optimal f'-repair and the m-set; the rest satisfies the FD under f' and is partitioned identically by f; then exchange the roles. The second: the kept values agree

iff no two kept cells $e \cap e'$ share an f-class or an f'-class, i.e. the kept cells form a matching, and keeping a cell entirely is never worse. □

7 Why not the pair-conditional measure

Let $g_1^c(X \to Y) = \Pr[t_1[Y] \neq t_2[Y] \mid t_1[X] = t_2[X],\ t_1 \neq t_2]$ over ordered pairs of distinct tuples.

Proposition 6. *There are instances with $g_1^c(X \to Y) = 0$, $g_1^c(Y \to Z) \leq 2k/m$ and $g_1^c(X \to Z) = 1$ for any $m > k \geq 2$, k even.*

Proof. Use one Y-value y. Add $m - k$ tuples (x_i, y, z_0) with distinct x_i, and for $j = 1..k/2$ two tuples (p_j, y, z_j^a), (p_j, y, z_j^b) with fresh values.

- All tuples share Y, so no X-agreeing pair disagrees on Y: $g_1^c(X \to Y) = 0$.
- Among the $m(m-1)$ ordered Y-agreeing pairs, the Z-agreeing ones are at least the $(m-k)(m-k-1)$ pairs within the z_0 group, so $g_1^c(Y \to Z) \leq 1 - \frac{(m-k)(m-k-1)}{m(m-1)} \leq \frac{2k}{m}$.
- The only X-agreeing pairs are the p_j pairs, which disagree on Z: $g_1^c(X \to Z) = 1$.

□

8 LLM oracle protocol (E8)

Model: Qwen2.5-7B-Instruct, Q4_K_M GGUF quantization (sha256 `65b8fcd9...a1423`), run with llama-cpp-python 0.3.36 on CPU (`n_ctx`=512, `temperature`=0, `max_tokens`=24, `seed`=0). For each normalized labeler value the model sees the most frequent raw spelling. The first line of the answer is lowercased and stripped of all non-alphanumeric characters, and the result is the entity label (so the oracle is a function). System prompt, verbatim:

> You normalize the names of companies that label drug products. Given one labeler name exactly as written in a product listing, reply with the canonical name of the company only: fix spelling and capitalization, drop legal suffixes (Inc, LLC, Ltd, Corp, Co), punctuation, store or division numbers, and 'dba'/'see also' remarks; if the listing names a store brand together with its owner, reply with the owner. Reply with the name only, no explanation.

Few-shot examples (synthetic firm names, given as prior chat turns):

- `user:` Acme Pharmaceuticals, Inc. → `assistant:` Acme Pharmaceuticals
- `user:` ACME PHARMACUETICALS LLC → `assistant:` Acme Pharmaceuticals
- `user:` Bestway Drug Co. (dba BestCare) → `assistant:` Bestway Drug
- `user:` Northwind Health 220, LLC → `assistant:` Northwind Health

All 6,653 labeler values were canonicalized once; the certification runs were replayed from these cached answers, each distinct queried value counting as one call. Code: `exp/e8_llm_oracle.py`.

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